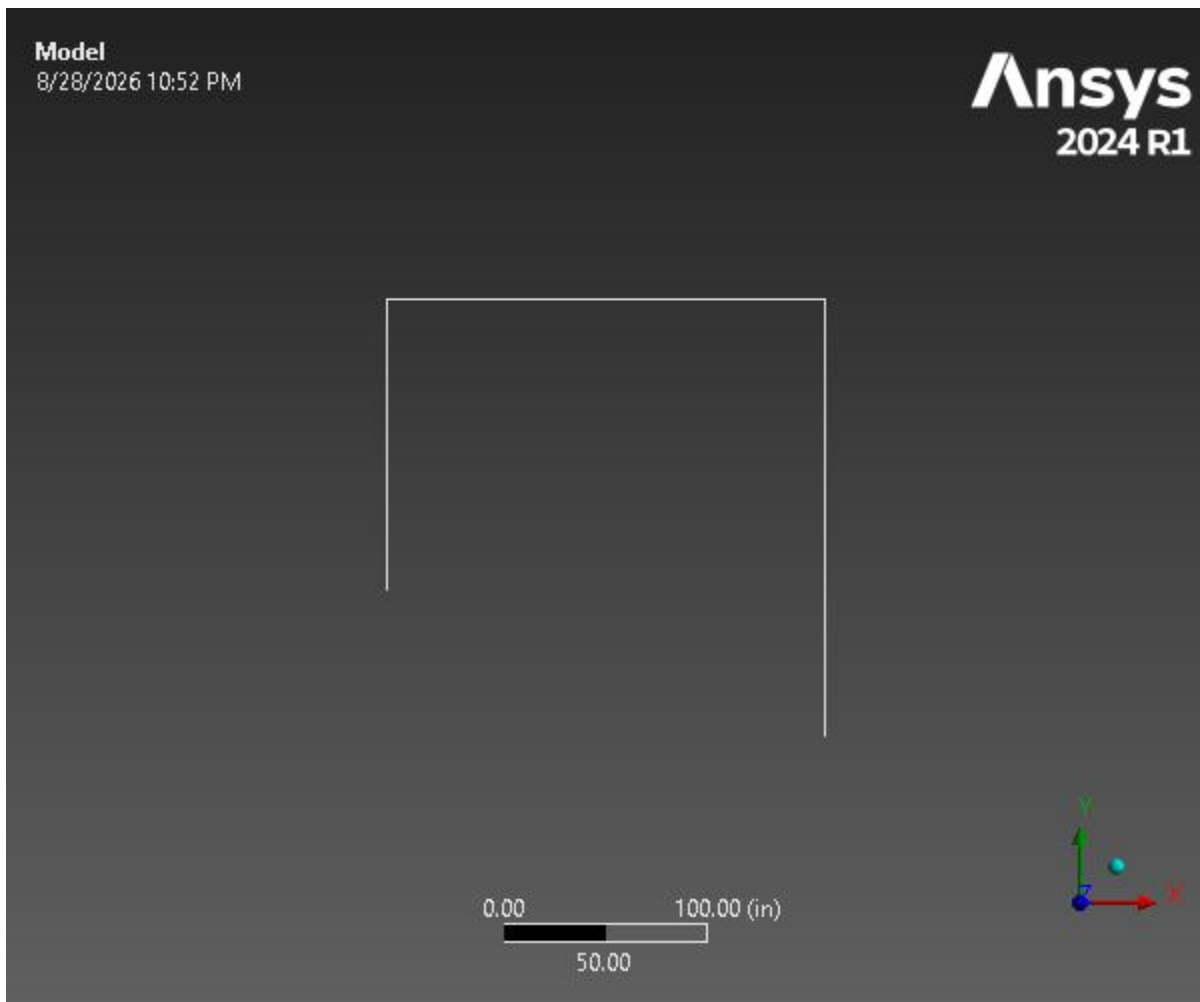




Project

First Saved	Friday, August 28, 2026
Last Saved	Friday, August 28, 2026
Product Version	2024 R1
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	U.S. Customary (in, lbm, lbf, s, V, A) Degrees rad/s Fahrenheit
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Fahrenheit

Model (B4)

TABLE 2

Model (B4) > Geometry Imports

Object Name	<i>Geometry Imports</i>
State	Solved

TABLE 3

Model (B4) > Geometry Imports > Geometry Import (A2, B3)

Object Name	<i>Geometry Import (A2, B3)</i>
State	Solved

Definition	
Source	C:\Users\Hp\AppData\Local\Temp\WB_Hp_32168_2\wbnew_files\dp0\Geom\DM\Geom.scdoc
Type	SpaceClaim
Basic Geometry Options	
Parameters	Independent
Parameter Key	
Advanced Geometry Options	
Compare Parts On Update	No
Analysis Type	3-D

Geometry

TABLE 4
Model (B4) > Geometry

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\Hp\AppData\Local\Temp\WB_Hp_32168_2\wbnew_files\dp0\Geom\DM\Geom.scdoc
Type	SpaceClaim
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	216. in
Length Y	216. in
Length Z	0. in
Properties	
Volume	18009 in ³
Mass	5107.2 lbm
Scale Factor Value	1.
Statistics	
Bodies	1
Active Bodies	1
Nodes	194
Elements	96
Mesh Metric	None
Update Options	
Assign Default Material	No

Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	Yes
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None

Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (B4) > Geometry > Parts

Object Name	<i>Beam (AISC_W18X106)</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Model Type	Beam
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Cross Section	AISC_W18X106
Offset Mode	Refresh on Update
Offset Type	Centroid
Treatment	None
Cross Section (For Solver)	Mesh
Material	
Assignment	Structural Steel
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	216. in
Length Y	216. in
Length Z	0. in
Properties	
Volume	18009 in ³
Mass	5107.2 lbm
Length	576. in
Cross Section Area	31.265 in ²
Cross Section IYY	1915.7 in ² ·in ²
Cross Section IZZ	220.47 in ² ·in ²
Statistics	
Nodes	194
Elements	96
Mesh Metric	None
CAD Attributes	
PartTolerance:	0.00000001
SCRootPartComponent	
Color:143.175.143	

TABLE 6
Model (B4) > Construction Geometry

Object Name	<i>Construction Geometry</i>
State	Fully Defined
Display	
Show Mesh	No

TABLE 7
Model (B4) > Construction Geometry > Paths

Object Name	<i>Path</i>
State	Fully Defined
Definition	
Path Type	Edge
Suppressed	No
Scope	
Scoping Method	Geometry Selection
Geometry	3 Edges

TABLE 8
Model (B4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	1
Material Assignments	0

TABLE 9
Model (B4) > Cross Sections

Object Name	<i>Cross Sections</i>
State	Fully Defined
Statistics	
Cross Sections	1

TABLE 10
Model (B4) > Cross Sections > AISC_W18X106

Object Name	<i>AISC_W18X106</i>
State	Fully Defined
Definition	
Type	MESH
Import Type	Imported
Physical Properties	
Beam Section	AISC_W18X106
A	31.265 in ²
I _{yy}	1915.7 in ² ·in ²
I _{zz}	220.47 in ² ·in ²
I _w	17234 in ⁶
J	7.9066 in ² ·in ²
CG _y	0. in
CG _z	9.35 in

SHy	-5.4845e-012 in
SHz	9.35 in

Coordinate Systems

TABLE 11
Model (B4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. in
Origin Y	0. in
Origin Z	0. in
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 12
Model (B4) > Connections

Object Name	<i>Connections</i>
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	0
Active Contacts	0
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 13
Model (B4) > Connections > Beam End Release

Object Name	<i>End Release</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Vertex Geometry	1 Vertex
Edge Geometry	1 Edge
Independent Edges	Yes
Definition	
Coordinate System	Global Coordinate System
Translation X	Fixed
Translation Y	Fixed
Translation Z	Fixed
Rotation X	Fixed
Rotation Y	Fixed
Rotation Z	Free
Behavior	Coupled
Suppressed	No

Mesh

TABLE 14
Model (B4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizing	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	305.47 in
Average Surface Area	0.0 in ²
Minimum Edge Length	144.0 in
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None

Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Batch Connections	
Mesh Based Connection	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Statistics	
Nodes	194
Elements	96
Show Detailed Statistics	No

TABLE 15
Model (B4) > Mesh > Mesh Controls

Object Name	<i>Edge Sizing</i>
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Geometry	3 Edges
Definition	
Suppressed	No
Type	Element Size
Element Size	6.0 in
Advanced	
Behavior	Soft
Bias Type	No Bias

Static Structural (B5)

TABLE 16
Model (B4) > Analysis

Object Name	<i>Static Structural (B5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural

Solver Target	Mechanical APDL
Options	
Environment Temperature	71.6 °F
Generate Input Only	No

TABLE 17
Model (B4) > Static Structural (B5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
State	Fully Defined
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes
Nonlinear Data	No
Nodal Forces	No
Volume and Energy	Yes

Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Engineering Portfolio\FEA Projects in Ansys\Frame Analysis\Frame Analysis_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	Bin

TABLE 18
Model (B4) > Static Structural (B5) > Loads

Object Name	Simply Supported	Fixed Support	Line Pressure	Line Pressure 2
State	Fully Defined			
Scope				
Scoping Method	Geometry Selection			
Geometry	1 Vertex		1 Edge	
Definition				
Type	Simply Supported	Fixed Support	Line Pressure	
Suppressed	No			
Define By			Components	
Coordinate System			Global Coordinate System	
X Component			100. lbf/in (ramped)	0. lbf/in (ramped)
Y Component			0. lbf/in (ramped)	-133.33 lbf/in (ramped)
Z Component			0. lbf/in (ramped)	

FIGURE 1
Model (B4) > Static Structural (B5) > Line Pressure

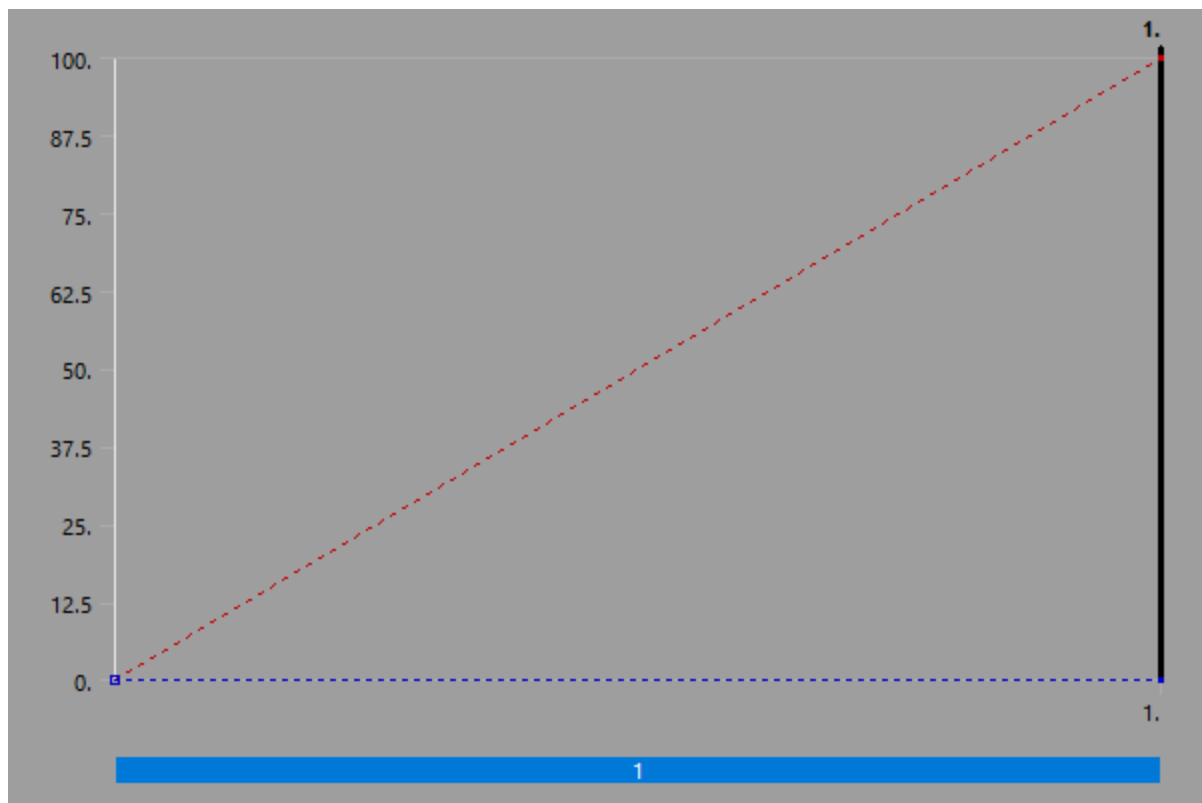
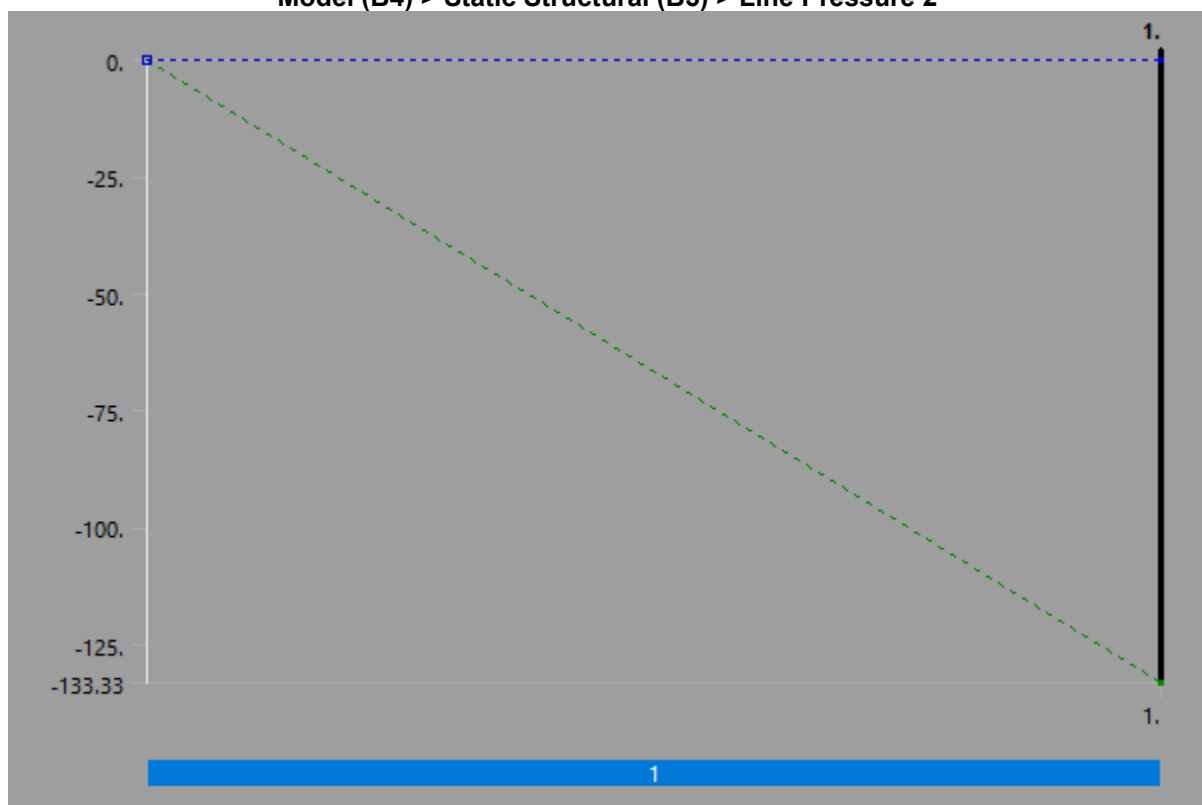


FIGURE 2
Model (B4) > Static Structural (B5) > Line Pressure 2



Solution (B6)

TABLE 19
Model (B4) > Static Structural (B5) > Solution

Object Name	<i>Solution (B6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	14. s
MAPDL Memory Used	207. MB
MAPDL Result File Size	1.25 MB
Post Processing	
Beam Section Results	Yes
On Demand Stress/Strain	No

TABLE 20
Model (B4) > Static Structural (B5) > Solution (B6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 21
Model (B4) > Static Structural (B5) > Solution (B6) > Results

Object Name	<i>Directional Deformation</i>	<i>Normal Elastic Strain</i>	<i>Axial Force</i>	<i>Total Bending Moment</i>	<i>Directional Shear-Moment Diagram (VZ-MY-UZ)</i>
State	Solved				
Scope					
Scoping Method	Geometry Selection				Path
Geometry	All Bodies		All Line Bodies		
Path					Path
Definition					

Type	Directional Deformation	Normal Elastic Strain	Directional Axial Force	Total Bending Moment	Directional Shear-Moment Diagram (VZ-MY-UZ)
Orientation	X Axis				
By	Time				
Display Time	Last				
Separate Data by Entity	No				
Coordinate System	Global Coordinate System		Solution Coordinate System		Solution Coordinate System
Calculate Time History	Yes				
Identifier					
Suppressed	No				
Graphics Display					Y Bending Moment
Results					
Minimum	-2.159e-006 in	-1.8893e-004 in/in	-16058 lbf	1.4082e-008 lbf·in	-6.7547e-009 lbf·in
Maximum	0.30857 in	1.5742e-004 in/in	-4713.7 lbf	1.0182e+006 lbf·in	
Average	0.19131 in	-1.2095e-005 in/in			
Minimum Occurs On	Beam (AISC_W18X106)				
Maximum Occurs On	Beam (AISC_W18X106)				
Information					
Time	1. s				
Load Step	1				
Substep	1				
Iteration Number	1				
Integration Point Results					
Display Option		Averaged	Unaveraged		
Average Across Bodies		No			
Graph Controls					
X-Axis					S

FIGURE 3
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation

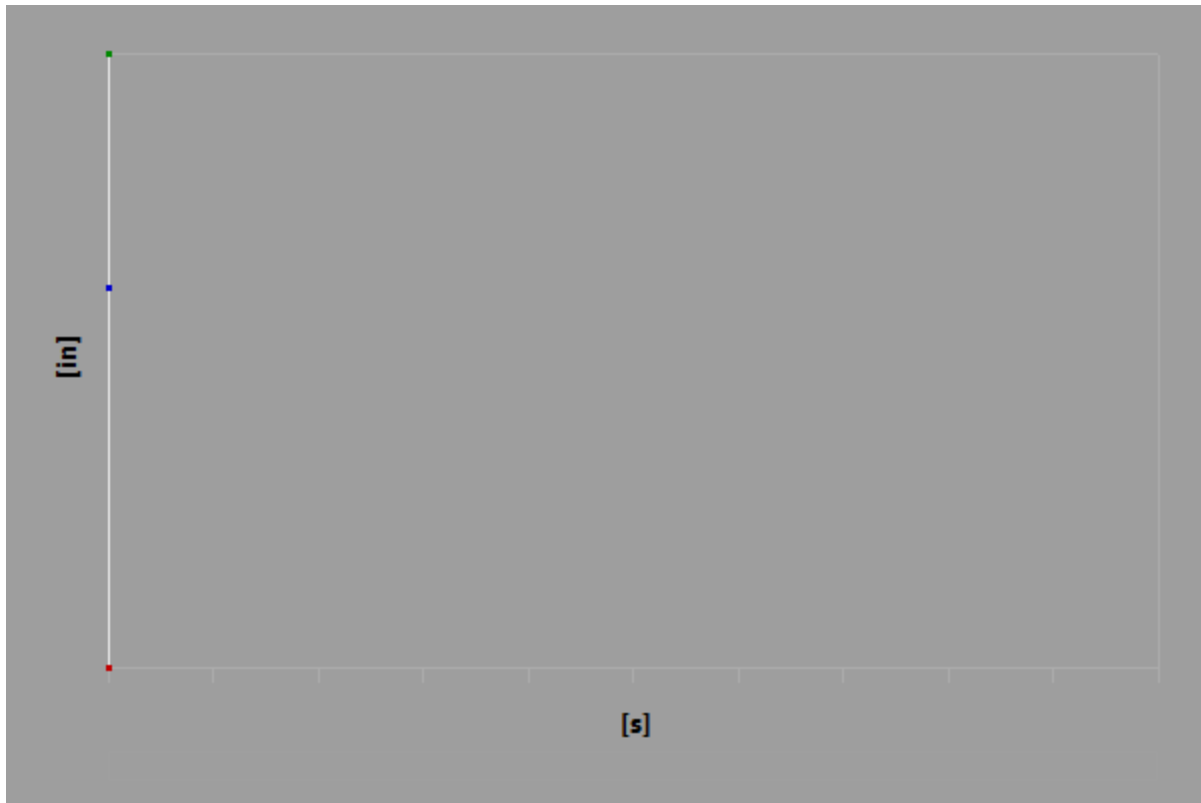


TABLE 22

Model (B4) > Static Structural (B5) > Solution (B6) > Directional Deformation

Time [s]	Minimum [in]	Maximum [in]	Average [in]
1.	-2.159e-006	0.30857	0.19131

FIGURE 4

Model (B4) > Static Structural (B5) > Solution (B6) > Normal Elastic Strain

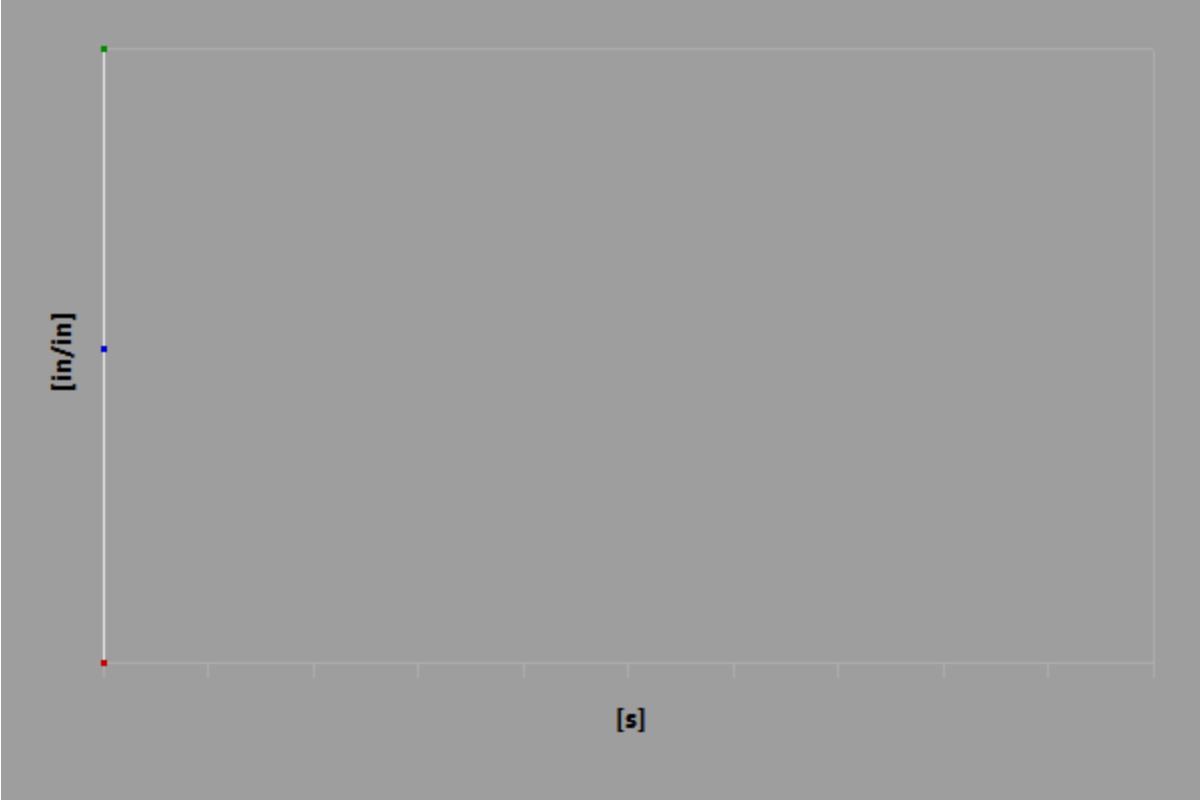


TABLE 23
Model (B4) > Static Structural (B5) > Solution (B6) > Normal Elastic Strain

Time [s]	Minimum [in/in]	Maximum [in/in]	Average [in/in]
1.	-1.8893e-004	1.5742e-004	-1.2095e-005

FIGURE 5
Model (B4) > Static Structural (B5) > Solution (B6) > Axial Force

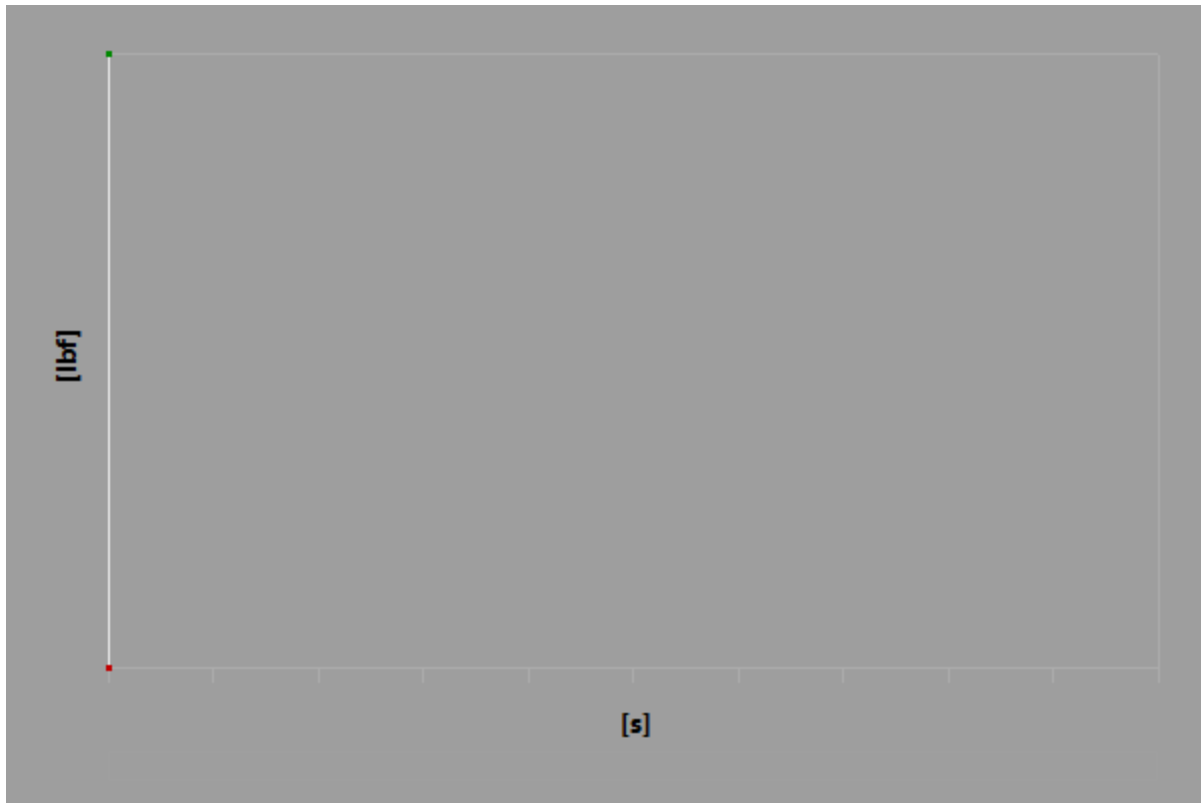


TABLE 24
Model (B4) > Static Structural (B5) > Solution (B6) > Axial Force

Time [s]	Minimum [lbf]	Maximum [lbf]
1.	-16058	-4713.7

FIGURE 6
Model (B4) > Static Structural (B5) > Solution (B6) > Total Bending Moment

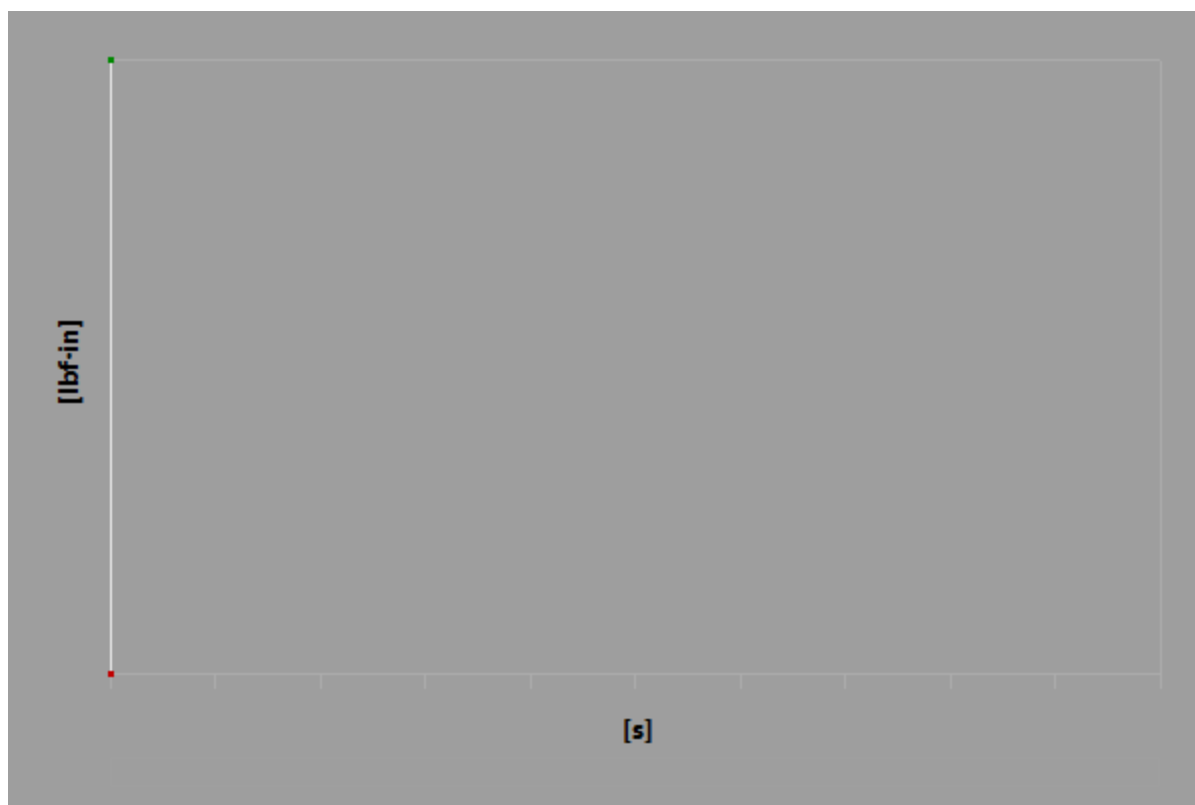


TABLE 25
Model (B4) > Static Structural (B5) > Solution (B6) > Total Bending Moment

Time [s]	Minimum [lbf·in]	Maximum [lbf·in]
1.	1.4082e-008	1.0182e+006

FIGURE 7
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Shear-Moment Diagram (VZ-MY-UZ)

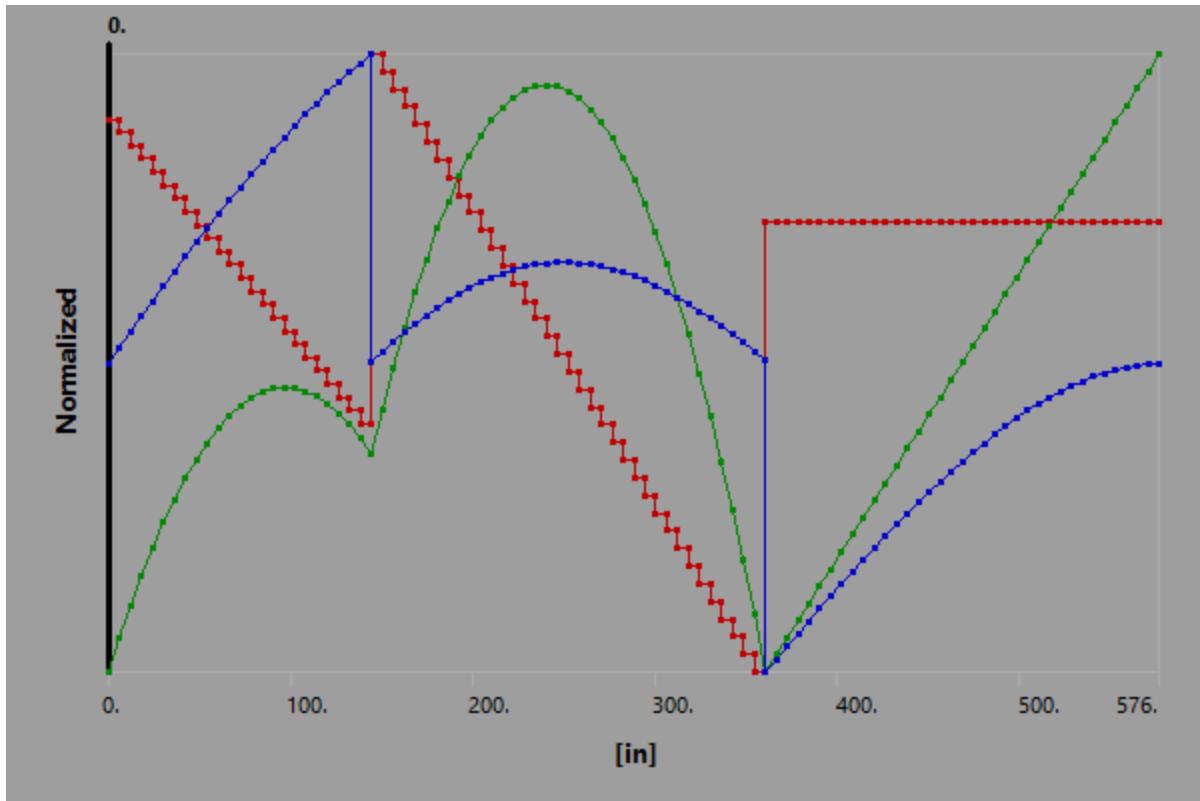


TABLE 26
Model (B4) > Static Structural (B5) > Solution (B6) > Directional Shear-Moment Diagram (VZ-MY-UZ)

Length [in]	Z Shear Force [lbf]	Y Bending Moment [lbf·in]	Z Displacement [in]
0.	9386.3	1.4086e-008	1.2093e-018
6.	8786.3	56318	1.4892e-002
12.	8186.3	1.0904e+005	2.9717e-002
18.	7586.3	1.5815e+005	4.4442e-002
24.	6986.3	2.0367e+005	5.9033e-002
30.	6386.3	2.4559e+005	7.3462e-002
36.	5786.3	2.8391e+005	8.7701e-002
42.	5186.3	3.1862e+005	0.10173
48.	4586.3	3.4974e+005	0.11551
54.	3986.3	3.7726e+005	0.12905
60.	3386.3	4.0118e+005	0.1423

66.		4.2149e+005	0.15527
	2786.3		
72.		4.3821e+005	0.16793
	2186.3		
78.		4.5133e+005	0.18028
	1586.3		
84.		4.6085e+005	0.19231
	986.29		
90.		4.6677e+005	0.20401
	386.29		
96.		4.6908e+005	0.21537
	-213.71		
102.		4.678e+005	0.2264
	-813.71		
108.		4.6292e+005	0.2371
	-1413.7		
114.		4.5444e+005	0.24747
	-2013.7		
120.		4.4235e+005	0.25751
	-2613.7		
126.		4.2667e+005	0.26723
	-3213.7		
132.		4.0739e+005	0.27665
	-3813.7		
138.		3.8451e+005	0.28578
	-4413.7		
144.		3.5803e+005	0.29462
	12342		2.0222e-003
150.		4.3208e+005	1.1487e-002
	11542		
156.		5.0133e+005	2.0631e-002
	10742		
162.		5.6579e+005	2.941e-002
	9942.5		
168.		6.2544e+005	3.7782e-002
	9142.5		
174.		6.803e+005	4.5709e-002
	8342.5		
180.		7.3035e+005	5.3154e-002
	7542.5		
186.		7.7561e+005	6.0086e-002
	6742.5		
192.		8.1606e+005	6.6475e-002
	5942.5		
198.		8.5172e+005	7.2295e-002
	5142.5		
204.		8.8257e+005	7.7523e-002
	4342.5		

210.		9.0863e+005	8.2139e-002
	3542.5		
216.		9.2988e+005	8.6126e-002
	2742.5		
222.		9.4634e+005	8.9471e-002
	1942.5		
228.		9.5799e+005	9.2162e-002
	1142.5		
234.		9.6485e+005	9.4192e-002
	342.48		
240.		9.669e+005	9.5557e-002
	-457.52		
246.		9.6416e+005	9.6255e-002
	-1257.5		
252.		9.5661e+005	9.6288e-002
	-2057.5		
258.		9.4427e+005	9.5662e-002
	-2857.5		
264.		9.2712e+005	9.4383e-002
	-3657.5		
270.		9.0518e+005	9.2464e-002
	-4457.5		
276.		8.7843e+005	8.9918e-002
	-5257.5		
282.		8.4689e+005	8.6762e-002
	-6057.5		
288.		8.1054e+005	8.3018e-002
	-6857.5		
294.		7.694e+005	7.8708e-002
	-7657.5		
300.		7.2345e+005	7.3859e-002
	-8457.5		
306.		6.7271e+005	6.8501e-002
	-9257.5		
312.		6.1716e+005	6.2667e-002
	-10058		
318.		5.5682e+005	5.6393e-002
	-10858		
324.		4.9167e+005	4.9718e-002
	-11658		
330.		4.2173e+005	4.2684e-002
	-12458		
336.		3.4698e+005	3.5336e-002
	-13258		
342.		2.6744e+005	2.7723e-002
	-14058		
348.		1.8309e+005	1.9896e-002
	-14858		

354.	-15658	93945	1.191e-002
360.		3.9524e-009	3.8225e-003
366.	4713.7	-6.7547e-009	-0.2935
372.		28282	-0.28139
378.		56565	-0.26931
384.		84847	-0.25726
390.		1.1313e+005	-0.24526
396.		1.4141e+005	-0.23334
402.		1.6969e+005	-0.22151
408.		1.9798e+005	-0.20979
414.		2.2626e+005	-0.1982
420.		2.5454e+005	-0.18675
426.		2.8282e+005	-0.17547
432.		3.1111e+005	-0.16437
438.		3.3939e+005	-0.15348
444.		3.6767e+005	-0.1428
450.		3.9595e+005	-0.13236
456.		4.2423e+005	-0.12218
462.		4.5252e+005	-0.11227
468.		4.808e+005	-0.10266
474.		5.0908e+005	-9.3356e-002
480.		5.3736e+005	-8.4382e-002
486.		5.6565e+005	-7.5756e-002
492.		5.9393e+005	-6.7497e-002
498.		6.2221e+005	-5.9622e-002
504.		6.5049e+005	-5.2149e-002
510.		6.7877e+005	-4.5098e-002
516.		7.0706e+005	-3.8487e-002
522.		7.3534e+005	-3.2333e-002
528.		7.6362e+005	-2.6655e-002
534.		7.919e+005	-2.1472e-002
540.		8.2019e+005	-1.6801e-002
546.		8.4847e+005	-1.2661e-002
552.		8.7675e+005	-9.071e-003
558.		9.0503e+005	-6.0483e-003
564.		9.3332e+005	-3.6115e-003
570.		9.616e+005	-1.779e-003
576.		9.8988e+005	-5.6908e-004
		1.0182e+006	0.

TABLE 27
Model (B4) > Static Structural (B5) > Solution (B6) > Probes

Object Name	Force Reaction	Force Reaction 2	Moment Reaction
State	Solved		
Definition			
Type	Force Reaction		Moment Reaction
Location Method	Boundary Condition		

Boundary Condition	Simply Supported	Fixed Support	
Orientation	Global Coordinate System		
Suppressed	No		
Summation			Centroid
Options			
Result Selection	All		
Display Time	End Time		
Results			
X Axis	-9686.3 lbf	-4713.7 lbf	1.9303e-002 lbf·in
Y Axis	12742 lbf	16058 lbf	5.7909e-002 lbf·in
Z Axis	-2.681e-004 lbf	2.681e-004 lbf	1.0182e+006 lbf·in
Total	16006 lbf	16735 lbf	1.0182e+006 lbf·in
Maximum Value Over Time			
X Axis	-9686.3 lbf	-4713.7 lbf	1.9303e-002 lbf·in
Y Axis	12742 lbf	16058 lbf	5.7909e-002 lbf·in
Z Axis	-2.681e-004 lbf	2.681e-004 lbf	1.0182e+006 lbf·in
Total	16006 lbf	16735 lbf	1.0182e+006 lbf·in
Minimum Value Over Time			
X Axis	-9686.3 lbf	-4713.7 lbf	1.9303e-002 lbf·in
Y Axis	12742 lbf	16058 lbf	5.7909e-002 lbf·in
Z Axis	-2.681e-004 lbf	2.681e-004 lbf	1.0182e+006 lbf·in
Total	16006 lbf	16735 lbf	1.0182e+006 lbf·in
Information			
Time	1. s		
Load Step	1		
Substep	1		
Iteration Number	1		

FIGURE 8
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction



TABLE 28
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction

Time [s]	Force Reaction (X) [lbf]	Force Reaction (Y) [lbf]	Force Reaction (Z) [lbf]	Force Reaction (Total) [lbf]
1.	-9686.3	12742	-2.681e-004	16006

FIGURE 9
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction 2

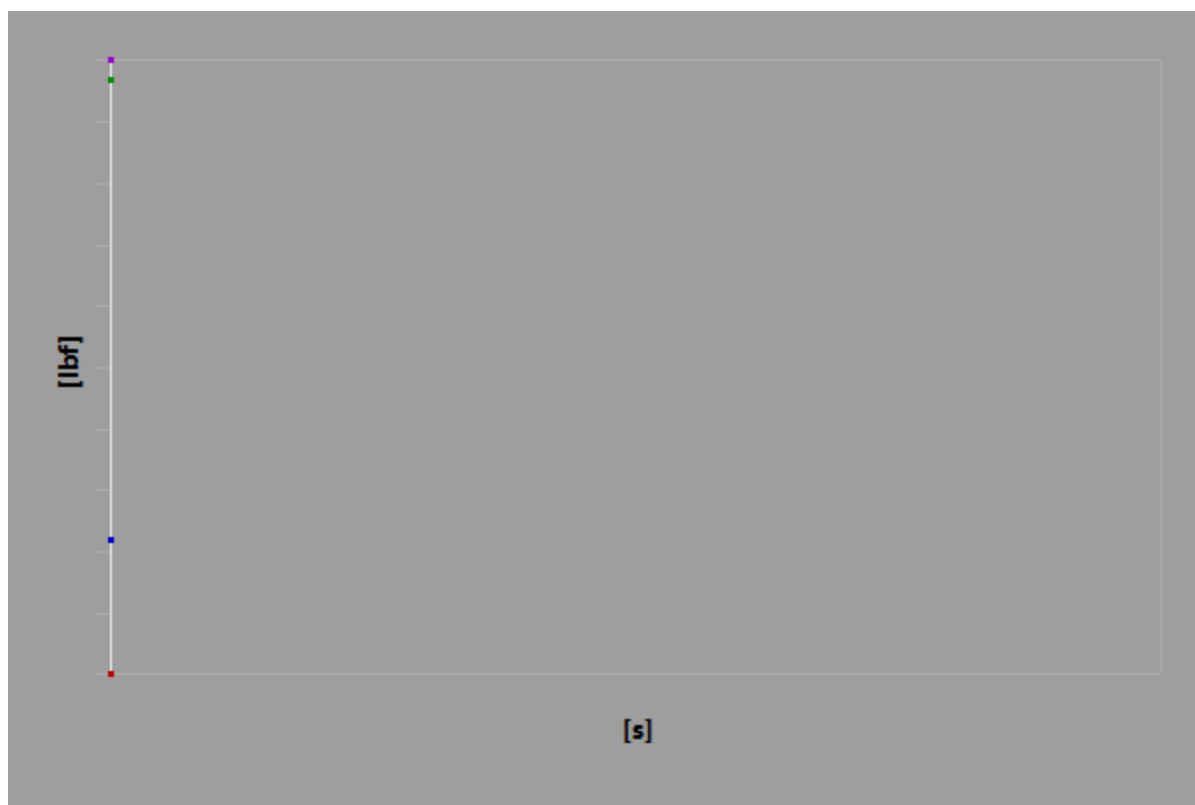


TABLE 29
Model (B4) > Static Structural (B5) > Solution (B6) > Force Reaction 2

Time [s]	Force Reaction 2 (X) [lbf]	Force Reaction 2 (Y) [lbf]	Force Reaction 2 (Z) [lbf]	Force Reaction 2 (Total) [lbf]
1.	-4713.7	16058	2.681e-004	16735

FIGURE 10
Model (B4) > Static Structural (B5) > Solution (B6) > Moment Reaction



TABLE 30
Model (B4) > Static Structural (B5) > Solution (B6) > Moment Reaction

Time [s]	Moment Reaction (X) [lbf·in]	Moment Reaction (Y) [lbf·in]	Moment Reaction (Z) [lbf·in]	Moment Reaction (Total) [lbf·in]
1.	1.9303e-002	5.7909e-002	1.0182e+006	1.0182e+006

Material Data

Structural Steel

TABLE 31
Structural Steel > Constants

Density	0.2836 lbm in ⁻³
Coefficient of Thermal Expansion	6.6667e-006 F ⁻¹
Specific Heat	0.10366 BTU lbm ⁻¹ F ⁻¹
Thermal Conductivity	8.0917e-004 BTU s ⁻¹ in ⁻¹ F ⁻¹
Resistivity	8.5235 ohm cmil in ⁻¹

TABLE 32
Structural Steel > Color

Red	Green	Blue
132	139	179

TABLE 33
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength psi
0

TABLE 34
Structural Steel > Compressive Yield Strength

Compressive Yield Strength psi
36259

TABLE 35
Structural Steel > Tensile Yield Strength

Tensile Yield Strength psi
36259

TABLE 36
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength psi
66717

TABLE 37
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature F
71.6

TABLE 38
Structural Steel > S-N Curve

Alternating Stress psi	Cycles	Mean Stress psi
5.8001e+005	10	0
4.1002e+005	20	0
2.7499e+005	50	0
2.0494e+005	100	0
1.5505e+005	200	0
63962	2000	0
38000	10000	0
31038	20000	0
20015	1.e+005	0
16534	2.e+005	0
12502	1.e+006	0

TABLE 39
Structural Steel > Strain-Life Parameters

Strength Coefficient psi	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient psi	Cyclic Strain Hardening Exponent
1.3343e+005	-0.106	0.213	-0.47	1.4504e+005	0.2

TABLE 40
Structural Steel > Isotropic Elasticity

Young's Modulus psi	Poisson's Ratio	Bulk Modulus psi	Shear Modulus psi	Temperature F
2.9008e+007	0.3	2.4173e+007	1.1157e+007	

TABLE 41
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000